KL UNIVERSITY • BACHUPALLY CAMPUS, HYDERABAD

Department of Computer Science & Engineering

MOVIE RECOMMENDATION ENGINE

Existing movie recommendation systems treat every viewer's taste as one-dimensional, rarely adapting to mood, language, or genre shifts.

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B.Tech II Year | Minor Project - Review 1 | Academic Year 2026-27 | Date: 18-09-2026

AGENDA

Contents

1 Introduction & Domain Background

2 Problem Statement & Motivation

3 Literature Survey

4 Research Gap Identified

5 Objectives of the Project

6 Proposed System Architecture

7 Methodology & Modules

8 Tools and Technologies

9 Work Completed / Demo

10 Timeline & Review Plan

11 Expected Outcome & Conclusion

12 References

SECTION 1

Introduction & Domain Background

What is the domain?

Domain: Movie Recommendation System

A movie recommendation system helps users discover movies that match their interests and preferences. It suggests personalized content based on what the user likes, using both content information (such as genre, cast, keywords, plot) and user ratings.

Why is it relevant today?

In 2025, with the rise of OTT platforms and global content, viewers have access to thousands of movies across multiple languages and cultures. Personalized recommendations save time, improve user satisfaction and help people find movies that truly match their mood and taste.

Where does your project sit?

This project uses machine learning techniques (content-based filtering with TF-IDF + cosine similarity and collaborative filtering with SVD) to recommend movies. It works with movie metadata (genre, cast, keywords, plot, language) and user ratings to generate accurate and diverse suggestions.



Panel tip

Every claim on this slide must be traceable to a reference on the last slide.

SECTION 2

Problem Statement & Motivation

PROBLEM STATEMENT

Existing movie recommendation systems fail to provide personalized recommendations because they mainly rely on popularity or basic search methods, which results in difficulty finding movies that match individual user preferences.

1 Who is affected?

Movie streaming users and viewers who have difficulty choosing movies that match their personal interests, genres, actors, and preferred storytelling styles.

2 What is the cost of the problem?

Time lost and reduced user satisfaction due to overwhelming choices and irrelevant recommendations, leading to decision fatigue.

3 Why solve it now?

The rapid growth of streaming platforms and the increasing number of available movies creates a need for personalized recommendations that can learn from user preferences and viewing/rating patterns.

SECTION 3

Literature Survey

S.No	Author & Year	Title / Technique Used	Dataset / Tools	Key Findings	Limitation / Gap
1	Sonika Malik (2022)	Movie Recommender System using Machine Learning – Content-Based & Collaborative Filtering, SVD, Cosine Similarity	Movie datasets; ML techniques	Combines content-based and collaborative filtering techniques for movie recommendations	Hybrid approach can be complex; scope for clearer comparison of individual approaches
2	Taushif Anwar & V. Uma (2021)	Comparative Study of Recommender System Approaches and Movie Recommendation Using Collaborative Filtering – SVD, SVD++, KNN, Co-clustering	Movie-rating data	Compares multiple recommendation approaches using collaborative filtering	Mainly focuses on collaborative filtering; limited focus on movie-content features
3	D. Phani Kumar et al. (2023)	Content Based Recommendation System on Movies – Count Vectorizer, Porter Stemmer & Cosine Similarity	Movie metadata	Uses movie-content features and cosine similarity to recommend similar movies	Focuses mainly on content-based filtering; does not provide SVD-based comparison
4	Yen-Liang Chen et al. (2021)	A Movie Recommendation Method Based on Users' Positive and Negative Profiles – Collaborative Filtering	User/movie rating data	Uses user preference profiles to generate personalized movie recommendations	Depends on user-rating information; cold-start and unrated-movie issues remain
5	Song Longsheng et al. (2022)	Collaborative Filtering Movie Recommendation Algorithm Based on SVD – SVD	Movie-rating data	Applies SVD-based collaborative filtering and addresses data sparsity	Mainly focuses on collaborative filtering; does not combine it with content-based similarity

SECTION 4

Research Gap Identified

GAP 1

Limited Integration of Data Existing studies often focus mainly on movie content such as genres, keywords, cast, and plot, or on user-rating patterns. Limited integration of both types of information can reduce the ability to capture different aspects of user preferences.

GAP 2

Many existing works concentrate on a single recommendation technique, such as content-based or collaborative filtering. There is a need for a clearer comparison of Cosine Similarity-based content filtering and SVD-based collaborative filtering.

GAP 3

Recommendation quality can be affected by limited user-rating information, data sparsity, and difficulty understanding individual preferences. These challenges can result in less personalized movie suggestions.

OUR PROJECT ADDRESSES

Existing work does not directly compare content-based and collaborative filtering side by side — our project builds both and evaluates them together to show their strengths and limitations.

SECTION 5

Objectives of the Project

1. **To collect / prepare** To collect and preprocess movie datasets such as MovieLens 100K and TMDB 5000, including movie metadata and user–movie ratings, for building the recommendation system.

2. **To design & implement** To design and implement Content-Based Filtering using Cosine Similarity and Collaborative Filtering using SVD to generate personalized movie recommendations.

3. **To evaluate** To evaluate and compare both recommendation approaches using suitable recommendation metrics and analyze their strengths, limitations, and recommendation quality.

4. **To deploy / demonstrate** To develop a working movie recommendation interface that presents personalized movie suggestions and allows users to explore the recommendations generated by both approaches.

SECTION 6

Proposed System Architecture

INPUT

MovieLens 100K + TMDB 5000

- User–movie ratings
- Movie metadata
- Genre, cast, keywords & plot

Raw movie & rating data

PRE-PROCESSING

Data Cleaning & Feature Preparation

- Handle missing data
- Process movie metadata
- Prepare rating matrix
- Extract movie features

Cleaned data & feature vectors

CORE MODULE

Movie Recommendation Engine

Content-Based Filtering
→ Cosine Similarity

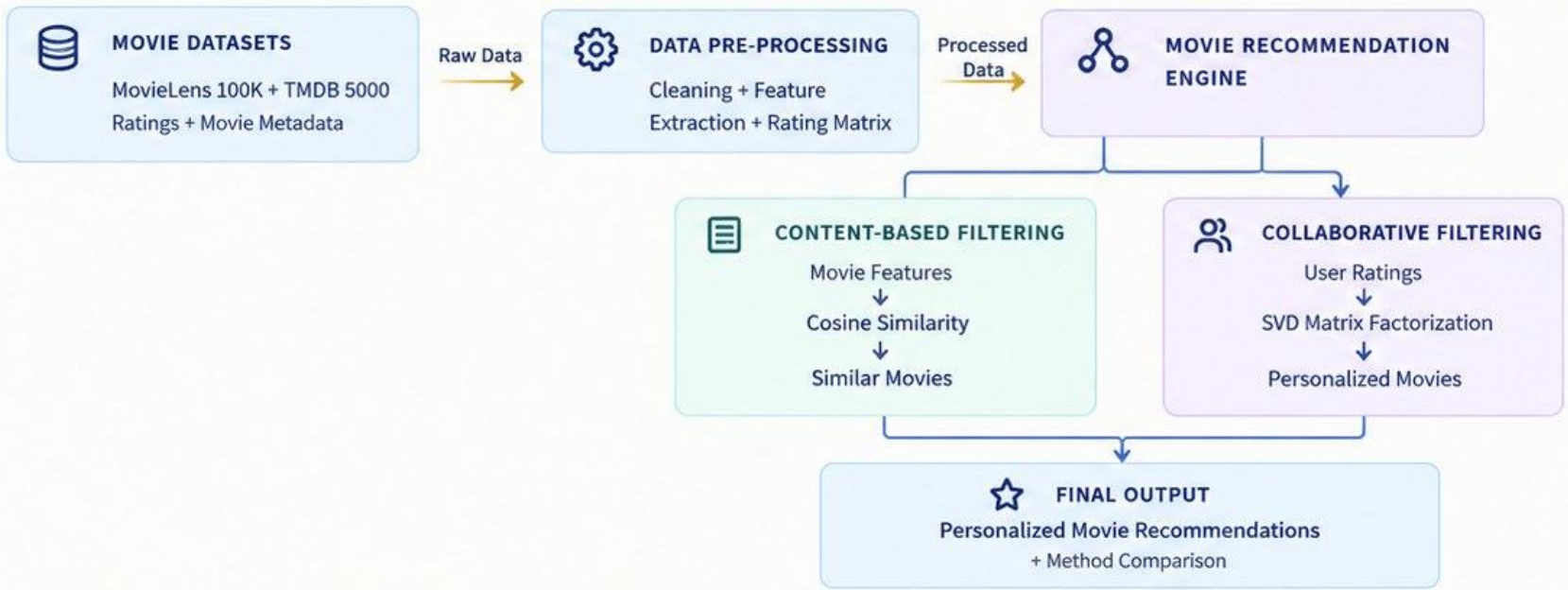
Collaborative Filtering
→ SVD Matrix Factorization

recommendation scores

OUTPUT

Personalized Movie Recommendations

- Similar movies
- Top-N recommendations
- Content vs. Collaborative results



RECOMMENDATION FLOW

Each stage passes processed movie data, feature vectors, or rating information to the next stage. The two recommendation methods generate results that are compared to provide personalized movie suggestions.

SECTION 7

Methodology & Modules

1 Module 1 - Data

- **Datasets:** MovieLens 100K and TMDB 5000
- **Attributes:** movie title, genres, overview/plot, cast, keywords, and user ratings
- **Source:** public movie datasets
- **Format:** CSV
- **Purpose:** build movie features and user–movie rating information
- **Data divided into** movie metadata and user-rating data

2 Module 2 - Processing

- **Data cleaning:** removed duplicate records and handled missing values
- **Text preprocessing:** combined movie overview, genres, cast, and keywords
- **Feature extraction:** TF-IDF vectorization for movie text features
- **Similarity calculation:** cosine similarity between movie feature vectors
- **Rating preparation:** created a user–movie rating matrix
- Prepared data for content-based filtering and SVD

3 Module 3 - Core Logic

- **Two recommendation approaches:** content-based and collaborative filtering
- **Content-based:** recommends movies similar to a selected movie
- **Technique:** TF-IDF + cosine similarity
- **Collaborative filtering:** uses user–movie rating patterns
- **Algorithm:** Matrix Factorization using SVD
- **Ranking and side-by-side** comparison of both methods

4 Module 4 - Interface

- **Frontend:** Streamlit web application (Python)
- **Backend:** Python using pandas, scikit-learn, and SVD-related libraries
- **User input:** movie selection/search and, if implemented, ratings
- System processes input and generates recommendations
- Displays movie titles, posters/details, and scores
- Shows both methods side by side for comparison

SECTION 8

Tools and Technologies

SOFTWARE

- Programming Language: Python 3.11+
- ML Libraries: Scikit-learn, Pandas, NumPy
- Library Recommendation: Cosine Similarity, TF-IDF, SVD (Matrix Factorization)
- Frontend / UI: Streamlit
- Data Format: CSV
- Development Environment: VS Code, Python Virtual Environment / Jupyter Notebook

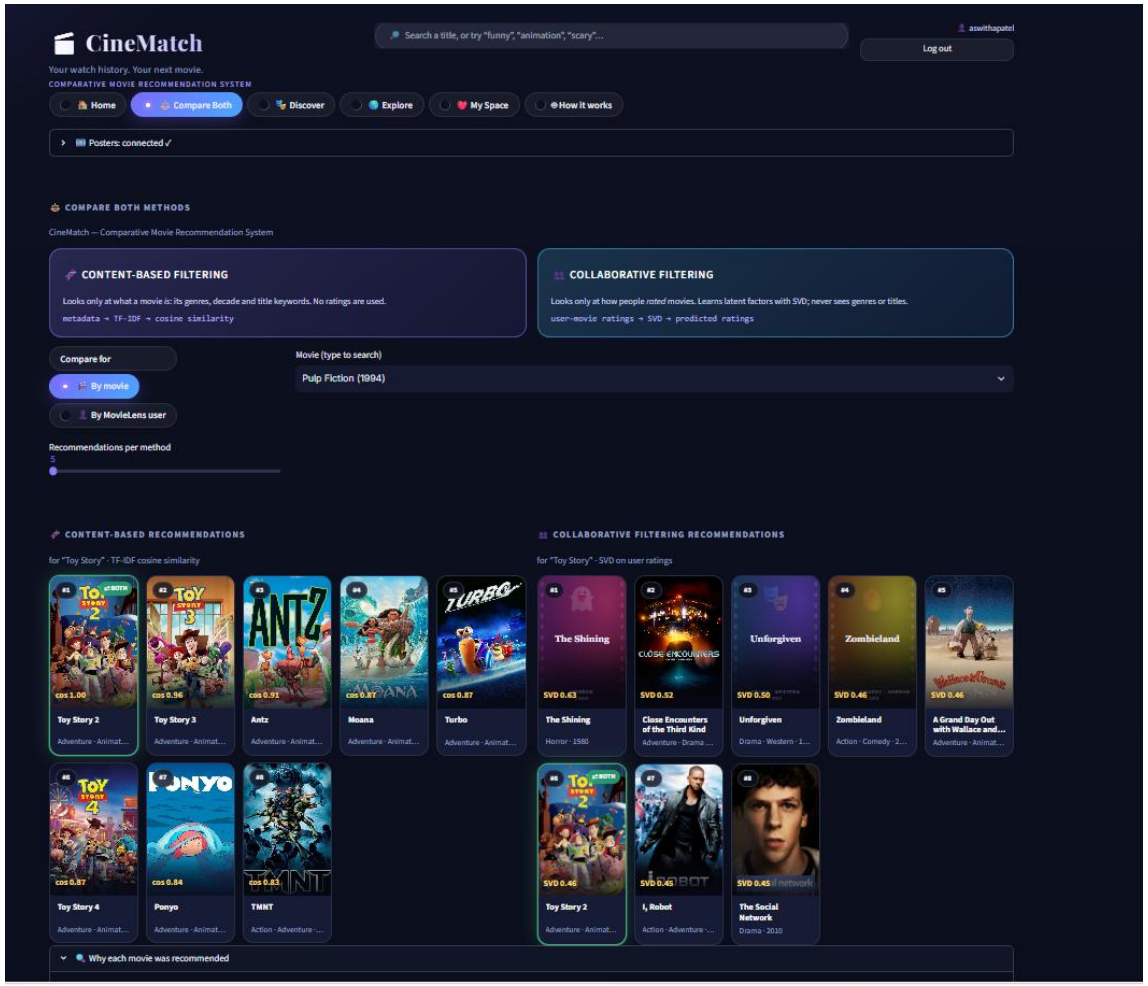
HARDWARE / PLATFORM

- Processor: Intel Core i5 or equivalent
- RAM: 8 GB or above
- GPU: Not required
- Operating System: Windows
- Platform: Localhost
- Deployment: Streamlit local server
- Internet: Required for fetching movie posters/metadata (TMDB API)

SECTION 9

Work Completed / Demo

Literature survey	Completed
Dataset collection	Completed
Pre-processing pipeline	In progress
Model training	In progress
User interface	Pending
Testing & evaluation	Pending



SECTION 10

Timeline & Review Plan

Review 0

Week 2

- Problem understanding and project scope
- Dataset selection plan (MovieLens, TMDB)
- Team roles finalised Initial proposal, abstract, and objectives

Review 1

Week 5

- Literature survey completed
- Dataset collection and exploratory analysis
- Pre-processing pipeline developed (cleaning, merging metadata)
- Methodology and model selection (content-based vs. collaborative filtering)

Review 2

Week 9

- System architecture finalised Model training and tuning (cosine similarity + SVD, 60% implementation)
- Streamlit interface development Initial testing and results comparison

Final

Week 14

- Complete system with both recommendation approaches
- Final testing and evaluation
- Full demo (movie recommendation with poster/metadata display)
- Final report and presentation

SECTION 11

Expected Outcome & Conclusion

DELIVERABLE

A working movie recommendation website that generates and displays personalized movie suggestions using content-based and collaborative filtering, with movie details and recommendation scores.

MEASURABLE TARGET

Generate relevant and ranked movie recommendations using TF-IDF, cosine similarity, and SVD, and compare the results of both approaches through the deployed website.

BEYOND THE LAB

The system can support personalized movie discovery through a web-based Streamlit interface, helping users explore movies based on content similarity and user-rating patterns.

CONCLUSION

Our project aims to solve the problem of finding relevant movies through personalized recommendations using content-based and collaborative filtering. The identified gap is the lack of a system that compares both approaches together; our system will deliver a working website with side-by-side recommendations using TF-IDF, cosine similarity, and SVD.

Future Scope: Real-time user feedback and continuously updated recommendations based on changing viewing history are left for future development.

SECTION 12

References

[1] Sonika Malik — *Movie Recommender System using Machine Learning* (2022)

[2] Taushif Anwar & V. Uma — *Comparative Study of Recommender System Approaches and Movie Recommendation Using Collaborative Filtering* (2021)

[3] D. Phani Kumar et al. — *Content Based Recommendation System on Movies* (2023)

[4] Yen-Liang Chen, Yi-Hsin Yeh & Man-Rong Ma — *A Movie Recommendation Method Based on Users' Positive and Negative Profiles* (2021)

[5] Vito Xituo Chen & Tiffany Y. Tang — *Incorporating Singular Value Decomposition in User-Based Collaborative Filtering Technique for a Movie Recommendation System: A Comparative Study* (2019).

This one is especially relevant because it directly discusses SVD and cosine similarity for movie recommendation.

Thank You

Questions & Suggestions

Team: 12 | Guide: Dr. G. Raghavendra | Dept. of CSE, KLH Bachupally